

CE 310

The Poisson Distribution

Week 4 · Class 1 · When Rare Events Have a Rate

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Part 1

When the Label Lies

"A 1,000-Year Flood"

October 4, 2015 · Columbia, South Carolina · 27 inches of rain in 48 hours

State officials and the National Weather Service labeled it a "**1,000-year flood event.**"

The implicit message: this happens once every thousand years. Extraordinary bad luck. Nothing to design around.

There was a problem.

Records showed that Columbia had experienced two **comparable flood events** in the previous six years.

Three "1,000-year" floods in six years.

If the 1,000-Year Label Were Correct...

A "1,000-year" event has $1/1000 = 0.001$ probability per year. Over **6 years**, the expected count is:

$$\lambda = 6 \text{ years} \times 0.001 / \text{year} = \mathbf{0.006} \text{ events in 6 years}$$

Under this model — accepting the label at face value — the probabilities are:

Event	Probability
$P(X = 0 \text{ in 6 years})$	99.40%
$P(X \geq 1 \text{ in 6 years})$	0.60%
$P(X \geq 3 \text{ in 6 years})$	< 0.0002%

The model says: in a world where the label is correct, virtually nothing should happen in 6 years.

But Columbia experienced three such events in six years.

What Three Events in Six Years Prove

Columbia observed three "1,000-year" events in six years. Two explanations:

Explanation	Assessment
Extraordinarily bad luck	< 1 in 500,000 — not a credible explanation
The label was wrong	Far more likely

💡 The Poisson model cannot say *why* the label was wrong — bad data, short record, or shifting climate. But it says the label **is** wrong, with near-certainty. This is what quantitative models do: they make wrong assumptions visible.

How a Wrong Label Gets Into the Standard

No one has observed a "1,000-year flood" directly — the label comes from extrapolating a statistical model well beyond the available data.

The core problem:

Columbia's gauge record spans ~80 years. Engineers fit a distribution to those observations, then extrapolated far into the tail to estimate the 0.1%-per-year flood level — and called it "1,000-year."

Projecting 12× beyond the record is like predicting a car will last 600,000 miles after driving it only 50,000.

How a Wrong Label Gets Into the Standard (cont.)

Two compounding errors:

- **Distribution choice:** Different statistical fits to the same 80 years of data produce very different 1,000-year estimates. There is no uniquely correct answer.
- **Non-stationarity:** A calibration from historical data assumes conditions haven't changed. Urbanization and shifting climate can silently raise the true flood rate.

⚠ These are not errors of negligence — they are limitations of the method. The Poisson model does not fix them; it reveals when the resulting label contradicts what we actually observed.

The Same Error Appears in Engineering

Design decision	Wrong approach	Consequence
Size a roof drain	Annual average rain rate	Undersized for monsoon bursts
Specify HVAC maintenance intervals	"Replace when it breaks"	Unplanned failures during peak season
Declare a storm return period	One data point, no model	Infrastructure fails at "impossible" events
Estimate bridge fatigue life	Mean traffic load	Actual failures exceed predictions

⚠ The common thread: using a **mean** or a **label** where a **distribution** is required. The average conceals the tail. The Poisson distribution makes the tail explicit.

The Question We Need to Answer

We know that rare events — rain days, equipment faults, extreme storms — occur at some **average rate** over time.

What we need:

- Not just *how often on average*, but *how likely is k events in a given period?*
- Not just *the mean*, but *the full distribution of counts*

What we're building today:

A probability model for **discrete counts of events** that occur at a known average rate — the **Poisson distribution**.

Part 2

The Poisson Model

Why We Need a Distribution

The mean ($\lambda = 3.86$ rain days/month for Tucson) tells you what is *typical*. It does not answer engineering questions.

Question	Mean alone	Full distribution
Expected rain days this month?	✓ 3.86	✓
P(exactly 8 rain days)?	✗	✓
P(more than 6 rain days)?	✗	✓
P(dry month — zero rain days)?	✗	✓

The **Probability Mass Function (PMF)** assigns a probability to every possible integer count k :

$$\text{PMF: } P(X = k) \quad \text{for } k = 0, 1, 2, \dots$$

For Poisson, the PMF is determined entirely by one parameter: λ . Every question in the right column is answered by evaluating the PMF at the appropriate k .

Poisson's Original Problem — and Why It Works

Siméon Denis Poisson, 1837: Original goal was modeling judicial errors — how often do juries wrongfully convict? He needed to count rare, independent events across a large number of trials.

The mathematical key — Poisson is the limit of the Binomial:

$$n \rightarrow \infty, \quad p \rightarrow 0, \quad n \times p = \lambda \text{ (constant)}$$

When any single event is rare but you have many chances for it, the Binomial simplifies to a one-parameter model. One number — λ — carries the entire distribution.

First engineering use: In 1898, Ladislaus Bortkiewicz counted deaths by horse kick in the Prussian cavalry — 14 corps over 20 years. Rare events, large exposure, constant average rate. The Poisson fit was near-perfect. The same three conditions now define every CE application you will use this week.

Three Conditions for Poisson to Apply

The Poisson distribution is appropriate when:

1. **You are counting events** — discrete occurrences in a fixed time or space (rain days per month, faults per year, calls per hour)
2. **Events are independent** — whether or not event k occurred does not affect the probability of event $k + 1$
3. **Average rate λ is constant** over the observation period

⚠ Poisson breaks down when events cluster (aftershocks following an earthquake), or when the rate changes systematically (monsoon season vs. winter). The solution: **condition first, then apply Poisson**. Seasonal λ values (July vs. Spring) handle the varying-rate problem exactly this way.

The PMF Formula

$$P(X = k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!} \quad k = 0, 1, 2, \dots$$

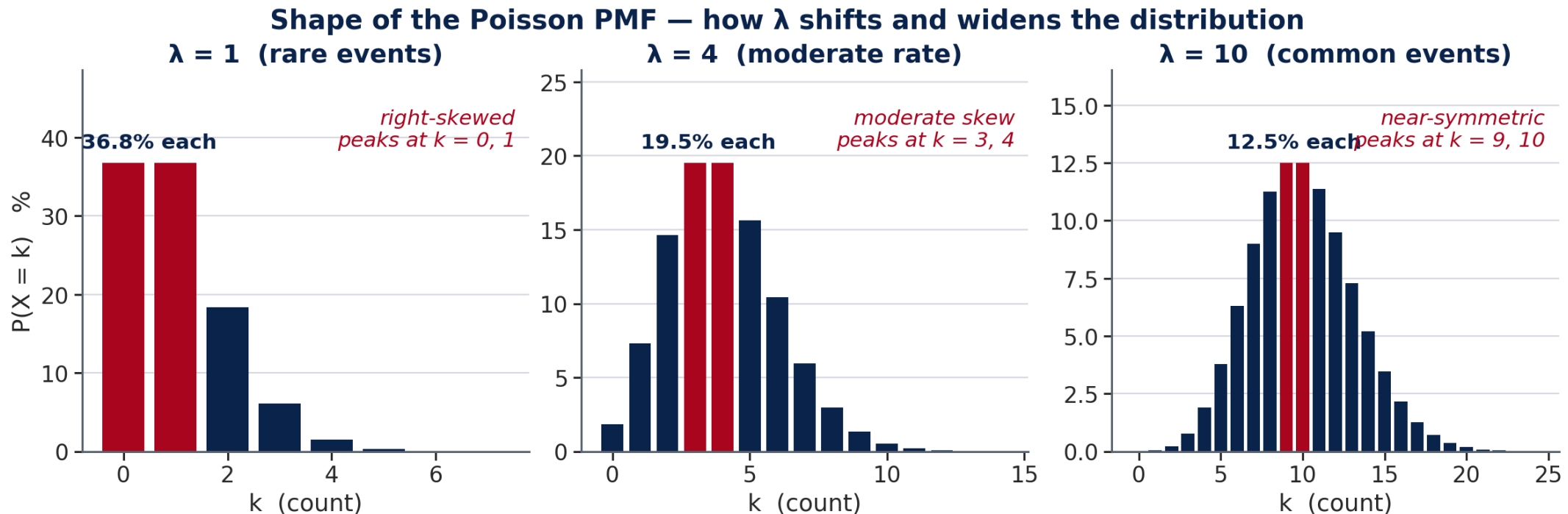
Each piece:

Symbol	Meaning
k	The count you're asking about (0, 1, 2, ...)
λ	The average rate (events per period)
$e^{-\lambda}$	Probability of zero events — the "baseline"
$\lambda^k / k!$	Relative weight of exactly k events

You do not need to compute this by hand in Excel — `=POISSON.DIST(k, λ, FALSE)` does it directly. But understanding the formula tells you *why* the distribution has the shape it does.

Shape of the PMF: How λ Changes Everything

Why the shape changes: At small λ , $P(X = 0) = e^{-\lambda}$ is large — zero events is the most probable outcome, creating strong right-skew. As λ grows, zero becomes rare, the mode shifts toward λ , and the distribution symmetrizes. Skewness = $1/\sqrt{\lambda}$ — it shrinks toward zero as λ grows, approaching the bell shape of the Normal distribution (Week 5).



What λ Tells You — Mean and Variance in One Number

For a Poisson distribution, λ is both the **mean** and the **variance**:

$$E[X] = \lambda \quad \text{Var}[X] = \lambda \quad \text{SD}[X] = \sqrt{\lambda}$$

💡 **Why equal?** Each event is an independent rare occurrence — the count's mean and variance both trace back to the same rate λ . Compare the Normal distribution (Week 5), where mean and variance are separate free parameters. In Poisson, one number controls everything.

Tucson overall: $\lambda = 3.86$ rain days/month

- Mean: 3.86 rain days
- Std deviation: $\sqrt{3.86} \approx \mathbf{1.96}$ rain days
- A month with 6 rain days is ≈ 1 SD above average

July monsoon: $\lambda = 10.3$ rain days/July

- Mean: 10.3 rain days
- Std deviation: $\sqrt{10.3} \approx \mathbf{3.21}$ rain days
- A July with 17 rain days is ≈ 2 SD above average

Answering the Week 3 Teaser

At the end of Week 3 we asked:

"If extreme-cooling hours occur at $\lambda = 4$ per week during July, what is $P(\text{exactly 6 such hours in a week})$?"

$$P(X = 6 \mid \lambda = 4) = \frac{e^{-4} \cdot 4^6}{6!} = \frac{0.01832 \times 4096}{720} \approx \mathbf{10.42\%}$$

In Excel: `=POISSON.DIST(6, 4, FALSE)` → **0.10420**

💡 10.42% — not negligible, not certain. The Poisson model says extreme-cooling weeks happen about one in ten times during July. An HVAC system designed only for the mean load ($\lambda = 4$ hours/week) will be undersized in roughly 40% of July weeks.

Quick Check — Part 2

Before we continue

Q: In a building with $\lambda = 2$ equipment faults per quarter, what is $P(\text{zero faults this quarter})$?

Use the formula: $P(X = 0) = e^{-\lambda}$

✓ Quick Check — Answer

$$P(X = 0 \mid \lambda = 2) = e^{-2} \approx 13.5\%$$

$$P(X = 0 \mid \lambda = 2) = e^{-2} = 0.1353$$

Approach	Result
Formula: $e^{-\lambda}$	$e^{-2} = 0.1353$
Excel: <code>=POISSON.DIST(0, 2, FALSE)</code>	0.1353

What this means: A building with $\lambda = 2$ equipment faults per quarter has about a **1-in-7 chance** of a completely fault-free quarter. Most semesters, at least one fault occurs.

💡 Students often overestimate this probability — they expect something closer to 50%. The exponential decay $e^{(-\lambda)}$ falls faster than intuition suggests.

Part 3

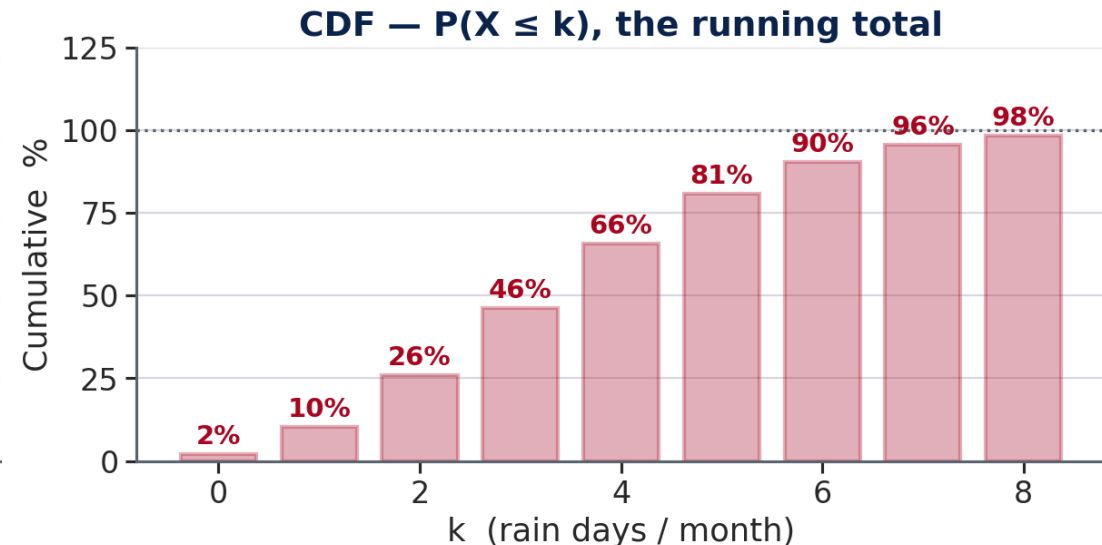
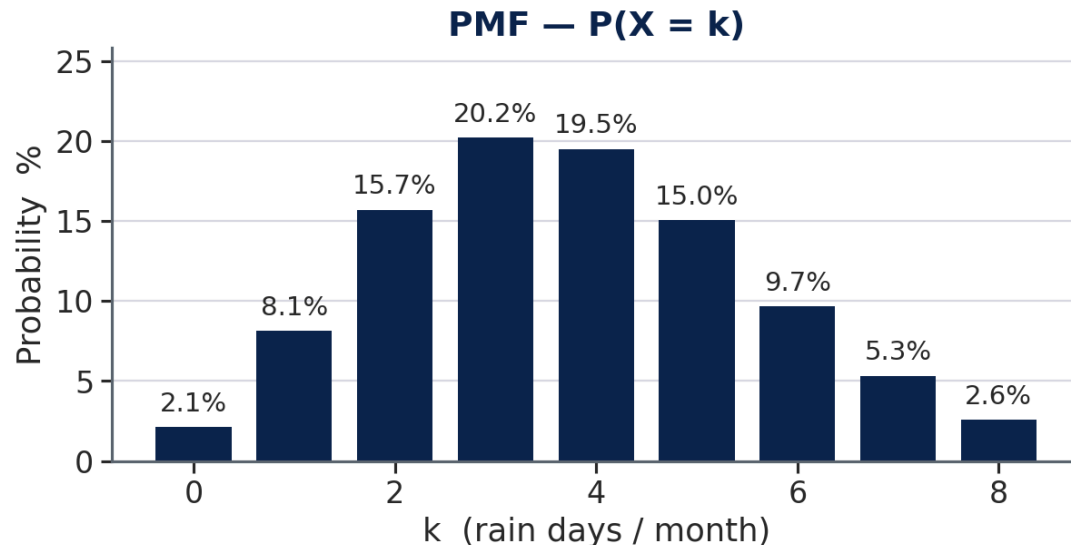
CDF and Exceedance Probability

From PMF to CDF

The **PMF** answers: $P(\text{exactly } k \text{ events})$? The **CDF** answers: $P(k \text{ or fewer events})$ — the running total of PMF values from 0 up to k . → See Tutorial for the Excel step-by-step

k	PMF	CDF
0	2.10%	2.10%
1	8.09%	10.19%

k	PMF	CDF
2	15.62%	25.80%
3	20.06%	45.87%



Exceedance: $P(X > k)$

How likely is the system to be stressed beyond design threshold k ?

$$P(X > k) = 1 - P(X \leq k) = 1 - \text{CDF}(k)$$

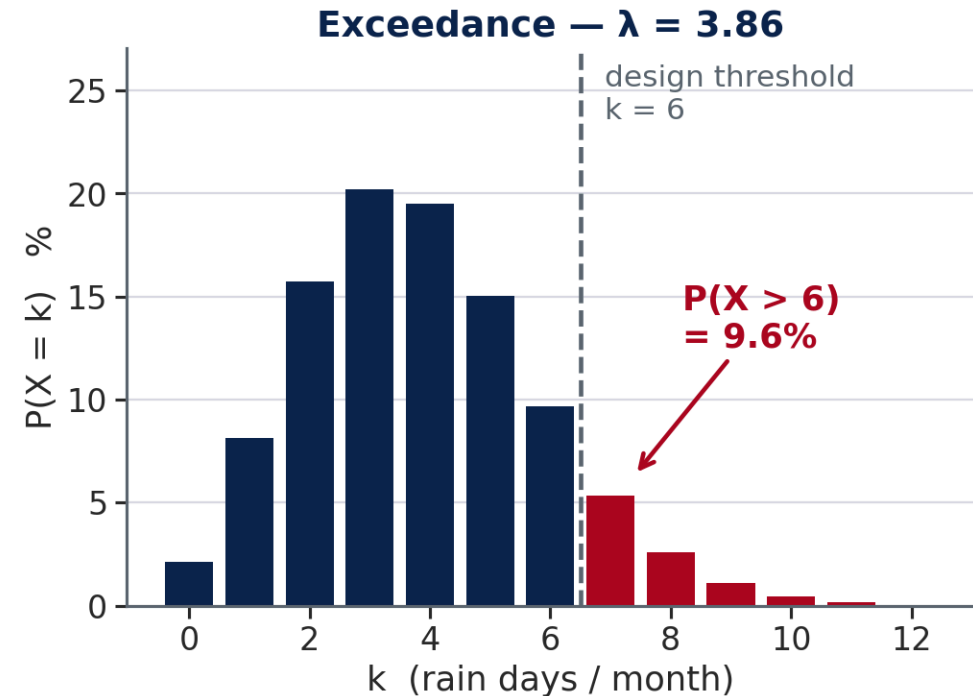
In Excel: `=1 - POISSON.DIST(k, λ, TRUE)`

Example: $\lambda = 3.86$ — more than 6 rain days?

$$P(X > 6 \mid \lambda = 3.86) = 1 - 0.9034 = \mathbf{9.66\%}$$



One in ten months at the overall rate.
Summer ($\lambda = 6.57$) raises this to ~48% — why seasonal conditioning matters.



Why Exceedance Is the Design Question

Design standard	Exceedance framing
Size roof drain: no more than 5% of months exceed capacity	Find k such that $P(X > k) = 5\%$
HVAC filter: replace when 90% of months are below limit	Find k such that $CDF(k) = 90\%$
Structural load: 2% in 50 years (ASCE 7 seismic)	$P(X \geq 1 \text{ in } 50 \text{ years}) = 2\%$

The question is never "what is the mean?" The question is always "what threshold ensures the tail probability is acceptable?"

Part 4

Return Periods and Design Life

What "Return Period" Actually Means

A **return period** T years means the event occurs at an average rate of $\lambda = 1/T$ **per year**.

- **100-year storm:** $\lambda = 0.01$ per year $\rightarrow P(\text{occurs in any given year}) = 1\%$
- **50-year storm:** $\lambda = 0.02$ per year $\rightarrow P(\text{occurs in any given year}) = 2\%$
- **1,000-year flood:** $\lambda = 0.001$ per year $\rightarrow P(\text{occurs in any given year}) = 0.1\%$

What return period does NOT mean: "this event will occur exactly once every T years." Poisson events cluster randomly — two 100-year storms can occur in back-to-back years.

⚠ Engineers who say "the 50-year storm won't happen this decade because it already occurred last year" are confusing a return period with a schedule. The clock does not reset. Each year is independent.

Predict — The 100-Year Storm

60 seconds · commit to a number before you compute

A box culvert is designed for the **100-year storm** — an event with a **1% chance in any given year**. The structure is expected to stay in service for a **50-year design life**.

What is the probability it sees at least one 100-year storm in that time?

- **A.** About 5%
- **B.** About 39%
- **C.** About 50%
- **D.** Essentially certain

Write your letter down. Do not compute — commit to an intuition first.

✓ 39.3% — Answer B

$$P(\text{at least one}) = 1 - e^{-t/T} = 1 - e^{-50/100} = 1 - e^{-0.5} = \mathbf{39.3\%}$$

Not 50%. The annual chances **do not add** — that would double-count every year in which more than one storm occurs.

💡 Read it the other way and it is more unsettling: there is a **60.7%** chance the culvert never sees the storm it was designed for — and a nearly 40% chance it does. Neither number is what "one-in-a-hundred" sounds like.

Probability Over a Design Life

The probability of at least one T -year event during an N -year project life:

$$P(\geq 1 \text{ event in } N \text{ years}) = 1 - e^{-N/T}$$

This is $P(X \geq 1)$ under Poisson with $\lambda = N/T$ events expected over the design life.

Return period T	10-year project	25-year project	50-year project
25-year storm	33.0%	63.2%	86.5%
50-year storm	18.1%	39.3%	63.2%
100-year storm	9.5%	22.1%	39.3%
500-year storm	2.0%	4.9%	9.5%

A 25-year storm has a 33% chance of occurring during a 10-year project — nearly one in three.

ASCE 7 and Design Return Periods

ASCE 7-22 (Minimum Design Loads for Buildings and Other Structures) specifies structural design loads using explicit exceedance probabilities over a design life:

Load type	Exceedance probability	Equivalent return period
Wind (Risk Category II)	7% in 50 years	≈ 700 years
Seismic (Risk Category II)	2% in 50 years	$\approx 2,475$ years
Flood (NFIP 1% annual chance)	1% per year	100 years

→ Tutorial walks through the calculation in Excel

This is exactly the formula you will use in lab cell B13 for a 25-year stormwater design event over a 10-year project life.

Poisson Counts Events — But Not How Bad They Are

Poisson answers: *How many* rain days occur in a month?

A structurally different question: *How intense is the worst storm of the year?*

Question	Random variable	Right tool
How many rain days this month?	Count of events	Poisson
What is the peak 24-hour flow rate?	Maximum of many draws	Extreme value
How strong is the worst wind gust this year?	Annual maximum	Extreme value

Why it matters for design: A drainage pipe fails when *peak flow* exceeds capacity — not when the count of rain days is high.

💡 Poisson models the *frequency* of events crossing a threshold — when the question shifts to *magnitude of the worst case*, a different distribution family applies.

A Preview: Extreme Value Distributions

When the question shifts from *how many events* to *how extreme was the worst one*, a different model family applies.

Poisson answers...	Extreme Value answers...
How many rain days this month?	How intense was the worst storm this year?
Count of threshold crossings	Magnitude of the annual maximum

ASCE 7 wind and flood loads come from fitting extreme value distributions to records of annual maxima — the return-period logic from Part 4 still applies, but the distribution family is different. The most common form is the **Gumbel distribution** (Type I Extreme Value), covered fully in Week 11.

 "How many" (Poisson) and "how extreme" (extreme value) are different questions requiring different tools. Week 11 goes deep.

Part 5

Estimating λ from Data

λ from Data: Count Events, Divide by Periods

$$\hat{\lambda} = \frac{\text{total events observed}}{\text{total observation periods}}$$

This is the **maximum likelihood estimator** for λ — the simplest and most natural estimate.

Example — Tucson precipitation, 2013–2022:

Calculation	Value
Total rain days in dataset	463
Total months in dataset	120 (10 years × 12)
$\hat{\lambda}_{\text{overall}} = 463 / 120$	3.858 rain days/month

💡 This single number — 3.858 — is both the mean and the variance of the monthly rain-day distribution.

Conditioning Changes λ , Not Just the Probability

In Week 3 we conditioned on season to compute $P(\text{event} \mid \text{context})$. Now we see that conditioning on season changes the **entire distribution** — it changes λ itself.

Season	Rain days (2013–2022)	Months	$\hat{\lambda}$
Winter (DJF)	117	30	3.90
Spring (MAM)	42	30	1.40
Summer (JJA)	197	30	6.57
Fall (SON)	107	30	3.57

Pooling all seasons ($\lambda = 3.858$) conceals that Summer has **4.7× the rate** of Spring. Any design based on the annual average is wrong for both seasons simultaneously — too conservative for spring, dangerously low for summer.

Predict — Rank the Seasons

60 seconds · rank before you look

Tucson rain days, 2013–2022. Four seasonal λ values, in rain days per month:

Rank these from wettest to driest — and estimate the ratio between the two extremes.

- Summer (Jun–Aug)
- Fall (Sep–Nov)
- Winter (Dec–Feb)
- Spring (Mar–May)

Then: which is more likely — a **summer** month with zero rain days, or a **spring** month with zero rain days? By roughly how much?

✓ Summer 6.57 · Winter 3.90 · Fall 3.57 · Spring 1.40

Season	λ (rain days/month)	$P(X = 0)$
Summer	6.5667	0.141%
Winter	3.9000	2.02%
Fall	3.5667	2.83%
Spring	1.4000	24.66%

Summer is **4.7×** **wetter** than Spring by rate. But the zero-rain probabilities differ by a factor of **175**.

💡 About **one spring month in four** sees no rain at all. For summer it is about **one month in 700**. The rate ratio is 4.7×; the consequence ratio is 175× — because the exponential in the PMF amplifies whatever λ does.

The Monsoon Signal: July vs. Spring

July: 103 rain days across 10 years $\rightarrow \lambda_{\text{july}} = 103 / 10 = \mathbf{10.3 \text{ rain days per July}}$

Spring month: $\lambda_{\text{spring}} = \mathbf{1.40 \text{ rain days per month}}$

Ratio: $10.3 / 1.40 = \mathbf{7.4\times}$ — July produces 7 times as many rain events as a spring month.

Question	Spring ($\lambda = 1.40$)	July ($\lambda = 10.3$)
P(zero rain days)	24.66%	< 0.01%
P(more than 5 rain days)	4.20%	95.17%
Std deviation	1.18 days	3.21 days

A July with zero rain days is essentially impossible. A spring month with zero rain days occurs almost one in four months. The same Poisson model captures both — with different λ .

Why Seasonal λ Matters for Drainage Design

Wrong: Design drain for the annual average (3.858 days/month).

Right (ASHRAE 90.1): Size for the peak monsoon month, not the annual average.

Design basis	P(drain undersized in July)
Annual average ($\lambda = 3.858$)	High — ignores monsoon completely
Summer average ($\lambda = 6.57$)	Lower — captures summer regime
July rate ($\lambda = 10.3$)	Lowest — sized for actual peak

Principle: condition before you model — use the λ for the operating context.

Part 6

Today's Lab

What You'll Do Today

Dataset: `Week4_Tucson_Precipitation_Daily_2013_2022.csv` — 3,652 daily rows (2013–2022)

Section A — Marginal Event Rate

- A1: Total rain days in the dataset
- A2: Overall λ (463 rain days / 120 months)
- A3–A5: PMF at $k=0$, PMF at $k=5$, CDF at $k=5$ (overall λ)
- A6–A8: Summer rain days, Summer λ , $P(X=10 \mid \text{Summer})$

Section B — Seasonal Poisson Analysis

- B1–B4: Spring rain days, Spring λ , $P(\text{dry spring month})$, $P(\text{wet spring month})$
- B5–B8: July rain days, July λ , $P(\text{dry July})$, $P(\text{extreme July} > 15 \text{ days})$

In-Lab Target and Take-Home

Chart 1 (in class): Clustered bar chart comparing the Spring PMF and Summer PMF for $k = 0$ to 15.

- Use `=POISSON.DIST(k_cell, $B$2, FALSE)` and `=POISSON.DIST(k_cell, $A$7, FALSE)`
- Lock λ with \$ signs; let the k cell shift as you fill down

Take-home: B7–B14, Chart 2 (seasonal λ bar chart), Memo PDF

⚠ B13 uses `=1 - EXP(-N/T)`, not `POISSON.DIST`. The `EXP` formula is the return-period formula derived from the Poisson CDF at $k=0$ — writing it with `EXP` makes the exponential structure explicit and matches engineering practice.

Looking Ahead — Week 5 Preview

Your PMF table for Summer ($\lambda = 6.57$) is bell-shaped, centered near $k = 6-7$.

Your PMF table for July ($\lambda = 10.3$) is wider and centered near $k = 10$.

The question to sit with: What happens to the shape of a Poisson PMF as λ grows very large — say, $\lambda = 50$ or $\lambda = 100$? Which continuous distribution does it approach?

💡 Week 5 is the **Normal distribution**. At large λ , $\text{Poisson}(\lambda) \approx \text{Normal}(\mu=\lambda, \sigma^2=\lambda)$. The discrete count distribution we built today converges to the continuous bell curve — and the tools generalize: instead of computing $P(X=k)$ for integers, we will compute $P(a \leq X \leq b)$ for any real interval.

Summary — Week 4

Rare events have rates — and rates have distributions

Today's concepts:

- **Poisson model** — counts of independent events at constant rate λ
- **PMF** — $P(X = k) = e^{-\lambda} \lambda^k / k!$ · In Excel: `POISSON.DIST(k,  $\lambda$ , FALSE)`
- **CDF** — $P(X \leq k)$ · In Excel: `POISSON.DIST(k,  $\lambda$ , TRUE)`
- **Exceedance** — $P(X > k) = 1 - \text{CDF}(k)$
- **Return period** — $P(\geq 1 \text{ event in } N \text{ years}) = 1 - e^{-(N/T)}$
- **λ estimation** — count events, divide by periods; condition on season before modeling

The core insight: The mean (λ) is necessary but not sufficient — the full PMF reveals the tail that the mean conceals.

Key Formulas — PMF, CDF & Exceedance

Concept	Formula
Poisson PMF	$P(X = k) = e^{-\lambda} \lambda^k / k!$
Mean and variance	$E[X] = \text{Var}[X] = \lambda$

Key Formulas — Return Period & λ Estimation

Concept	Formula
λ from data	$\hat{\lambda} = \text{events/periods}$